Tech Interview Prep:

Architectural Styles & Event-Driven Systems

Section 1: Monolithic vs. Microservices Architectures

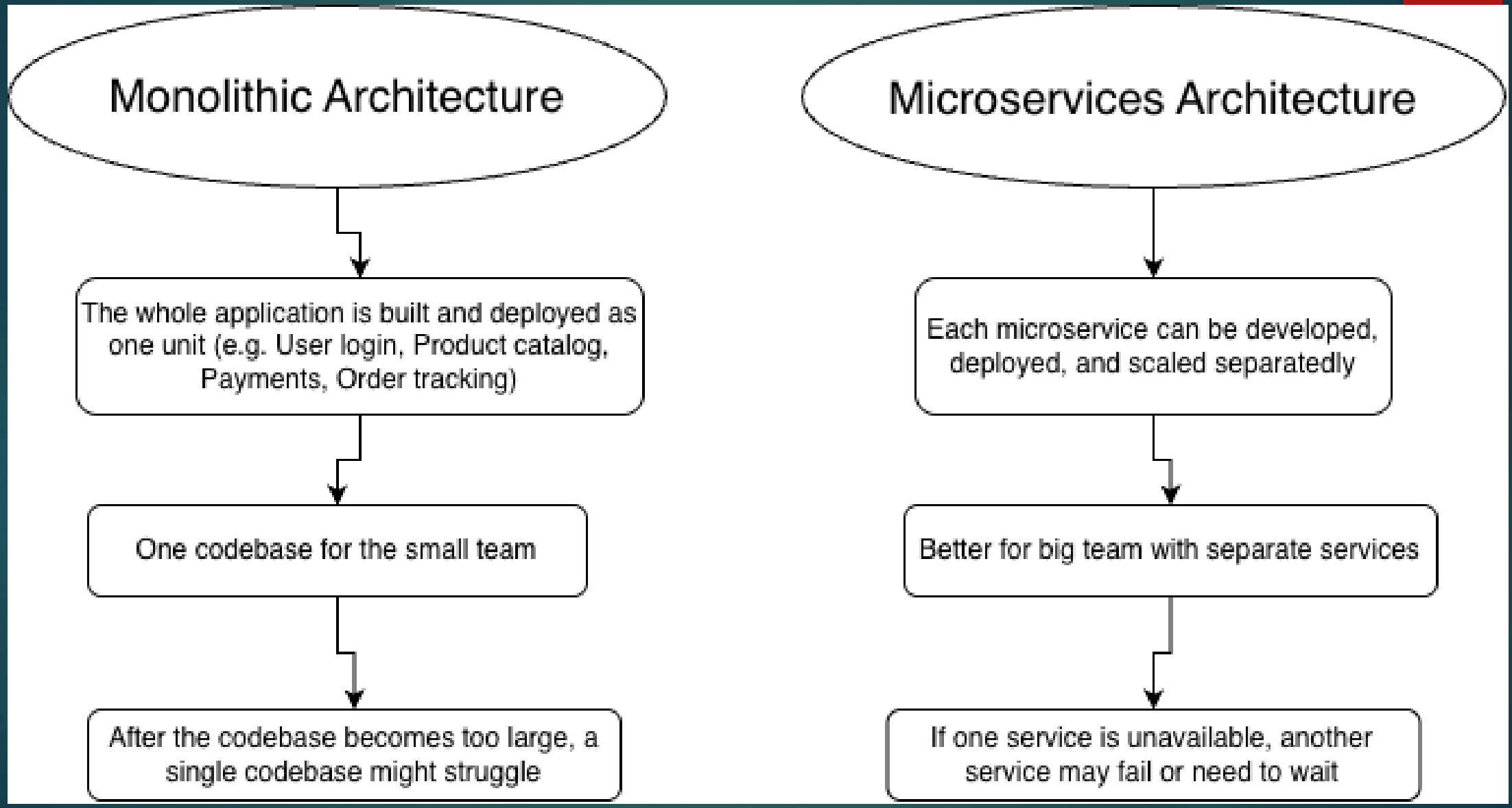
Section 2: Event-Driven Architecture (EDA) & Asynchronous Decoupling

Section 3: The Recommendation / Synthesis

NAME: DARA SEN

CLASS: AD371 (APPLICATION ARCHITECTURE)

Section 1: Monolithic vs. Microservices Architectures



Section 2: Event-Driven Architecture (EDA) & Asynchronous Decoupling

Event-Driven Architecture and Asynchronous Decoupling

A communication style that reacts when something happens (e.g. when there is an order, payment)

Event-driven architecture => asynchronous decoupling

Order servier sends a message =>

Inventory service: reduce stock
Email service: send receipt
Shipping service: prepare deliver
Analytics service: record the sale

asynchronous = because the Order service does not wait for other services service

decoupling = because the Order service does need to know or call those services

Section 3: The Recommendation / Synthesis

- Evolve architecture gradually (modular monolith → microservices)
- Move to microservices when there are actual needs, different teams
- Use event-driven architecture for work that does not need an immediate response (e.g. sending confirmation emails, updating loyalties points, etc.)